

Table 7. Illustrative out-of-sample WTP predictions

<i>Description</i>	Acute condition 1			Acute condition 2			Acute condition 3			Acute condition 4		
	1	10		1	10		1	10		1	10	
Duration ΔDAYS												
Severity: symptoms	Breathing smog or unpleasant air			Cough, wheezing, shortness of breath			Headache or dizziness			Sick, vomiting		
QWBSYSCORE	0.101	0.101		0.257	0.257		0.244	0.244		0.29	0.29	
Severity: functional restrictions												
	Moderate			Moderate			Severe			Severe		
QWBMOBSCORE	0.062	0.062		0.062	0.062		0.09	0.09		0.09	0.09	
QWBSACSCORE	0.06	0.06		0.06	0.06		0.077	0.077		0.077	0.077	
QWBPACSCORE	0.061	0.061		0.061	0.061		0.106	0.106		0.106	0.106	
Severity: total ΔQWB	0.284	0.284		0.44	0.44		0.517	0.517		0.563	0.563	
<i>Valuation</i>												
BT function 1 ^{a,b}												
95% CI Lower	\$38.78	\$129.94		\$88.59	\$264.25		\$109.01	\$318.47		\$119.68	\$345.19	
Predicted WTP/case avoided	\$44.94	\$142.51		\$125.12	\$396.73		\$182.43	\$578.49		\$222.68	\$706.10	
95% CI Upper	\$51.10	\$155.07		\$161.64	\$529.22		\$255.86	\$838.51		\$325.68	\$1067.02	
BT function 2 ^{a,b}												
95% CI Lower	\$67.54	\$227.59		\$100.26	\$257.74		\$110.91	\$263.63		\$102.08	\$226.85	
Predicted WTP/case avoided	\$124.11	\$371.86		\$158.32	\$474.39		\$218.80	\$655.58		\$229.83	\$688.66	
95% CI Upper	\$180.67	\$516.13		\$216.39	\$691.04		\$326.68	\$1047.53		\$357.58	\$1150.46	
QALY valuation estimate ^c	\$77.81	\$778.08		\$120.55	\$1205.48		\$141.64	\$1416.44		\$154.25	\$1542.47	

^a Confidence intervals calculated using the estimated asymptotic variance of the WTP functions.^b All WTP predictions assume the following values for the other explanatory variables: INCOME = \$45 000, %MALE = 50, AGE = 45, US = 1, WTPAVOID = 1, JOURNAL = 1.^c (ΔDAYS/365) (ΔQWB) × \$100 000.

estimate WTP for avoided illness. New primary research in health valuation is always desirable, but as this study shows, much can be learned from the existing body of empirical research.

The results of the meta-regression analysis indicate that WTP estimates for avoided acute effects vary in systematic and expected ways with respect to key explanatory variables. We find a strong statistical relationship between the value estimates and corresponding measures of the severity (using the QWB Scale) and duration of the health effects. We find that the average income and age of the study populations are statistically and economically significant explanatory factors in explaining WTP variation, which provides support for the theoretical validity of the stated preference approach. These results have also been confirmed by a recent study that expanded on the Johnson *et al.* meta-analysis with a somewhat different and less expanded sample of WTP estimates [37].

In addition, our meta-analysis results provide a simple but informative test of the assumptions underlying the QALY valuation approach for assessing morbidity values. The results strongly reject the assumption of a constant value per QALY and the assumption that the duration and the severity of illness have equivalent and proportional effects on WTP.

Finally, the results also indicate that the four health utility scores underlying the QWB Scale have statistically different effects on WTP. The mobility and physical activity dimensions were found to have stronger effects on WTP than the symptoms or social activity dimensions. This finding contradicts the equal weighting assumption, which is typically used in constructing the composite QWB health status index.

A key issue is therefore what these results imply for benefit transfer. First, the results cast doubt on the QALY valuation approach for predicting and transferring health benefits, at least for acute conditions. This result is consistent with the findings of others who have been critical of the approach, particularly for valuing changes in acute illness [38]. Second, the strong explanatory power, robustness, and statistical significance of the meta-regressions provide general support for their use as benefit transfer functions for policy analysis. In the absence of WTP studies that specifically match the policy effects of interest, these equations provide a systematic and empirically based approach for predicting WTP. BT Function 2 in particular provides a transfer tool that can differ-

entiate values according to the underlying dimensions of the health condition (i.e. symptoms, mobility, and activity).

Some caution is required, however, in interpreting and using the meta-analysis results. First, and most importantly, they only apply to acute conditions. Through our search of the health valuation literature, we determined that the number of WTP studies for chronic conditions is currently not adequate to support a comparable meta-analysis of these values. For chronic conditions, there are additional challenges in developing appropriate measures of duration and severity of illness. Second, the size of the estimated marginal effect of age on WTP in the meta-regressions, although statistically significant and reasonably robust across functional specifications, is unexpectedly high. Further investigation of the age effect on WTP is warranted, particularly as more WTP studies become available.

Third, it is important to emphasize that the meta-analysis function only includes estimates of adults' *private* WTP to avoid morbidity outcomes. As a result, it may not include all the value components that are of potential interest to policy makers. To the extent that costs of illness are externalized to other members of society, for example through social insurance, this analysis will not capture the benefits of avoiding these externalized costs. By excluding these externalized costs, estimates of *private* WTP may in some cases significantly underestimate the total social benefits of avoided illness. Moreover, the meta-analytic functions cannot be applied to *specifically* estimate values for protecting children's health because, by design, none of the included WTP estimates address altruistic (i.e. parental) values.

Fourth, it should be emphasized that the strength of these results depends in large part on the strength of the underlying WTP estimates and QWB scores. It is encouraging that the convergent validity of the two measures is supported by the finding that WTP is positively and significantly related to QWB scores. Nevertheless, the results would be weakened if either of these two preference measures includes systematic biases. Although the empirical accuracy and reliability of both methods have been questioned (see [39] for a critique of CVM methods and [16] for a review of QWB evaluations), there is no conclusive evidence that such systematic biases exist.

Table A1. Summary of studies included in the meta-analysis of WTP estimates^a

Study number	Study name	Valuation method	Health effect/change	Country	Number of WTP Obs
1	Alberini and Krupnick [40]	CVM	Acute respiratory symptoms	Taiwan	3
1	Alberini <i>et al.</i> [41]	CVM	Acute respiratory symptoms	Taiwan	2
2	Bala <i>et al.</i> [42]	CVM	Pain from shingles	US	1
3	Chestnut <i>et al.</i> [43]	CVM	Angina attacks	US	2
3	Rowe and Chestnut [44]	CVM	Asthma attacks	US	1
4	Dickie and Ulerly [45]	CVM	Acute symptoms	US	15
5	Johnson <i>et al.</i> [31]	Conjoint	Acute symptoms/activity restrictions	Canada	67
6	Kartman <i>et al.</i> [46]	CVM	Angina attacks	Sweden	12
7	Keith <i>et al.</i> [47]	CVM	Allergic rhinitis	US	1
8	Liu <i>et al.</i> [32]	CVM	Cold	Taiwan	1
9	Ready <i>et al.</i> [48]	CVM	Acute symptoms	Norway	10
9	Ready <i>et al.</i> [49]	CVM	Acute symptoms	Norway, Netherlands, Portugal, Spain, England	27
10	Navrud [50]	CVM	Acute symptoms	Norway	18
11	Lee <i>et al.</i> [51]	Conjoint	One day of influenza symptoms	US	3
12	Torrance <i>et al.</i> [52]	CVM	Acute exacerbation of chronic bronchitis	Canada	2
13	Jacobs <i>et al.</i> [53]	CVM	Hepatitis A	US	1
14	Chestnut <i>et al.</i> [54]	CVM	Angina attacks	US	2
15	Dickie <i>et al.</i>^b	CVM	Acute symptoms	US	12
16	Loehman <i>et al.</i> [55]	CVM	Acute symptoms	US	36
17	Tolley and Babcock [56]	CVM	Acute symptoms	US	20
				Total	236

^aPublications shown in bold font were included in the Johnson *et al.* [3] meta-analysis.^bDickie M, Gerking S, McClelland G, Schulze W. Contingent valuation: the value of formation process. Unpublished manuscript, 1988.

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Appendix A

The detailed descriptions of some of the key characteristics of these studies, including the valuation methods used, the health effects valued, the country studied, and the number of value estimates included in the meta-analysis are provided in Table A1.

References

1. Agency for Healthcare Research and Quality (AHRQ). Focus on cost-effectiveness analysis at the Agency for Healthcare Research and Quality. Fact Sheet. AHRQ Pub. No. 01-PO23, 2001.
2. Harvard School of Public Health (HSPH), Harvard Center for Risk Analysis. Catalog of preference scores. *The CEA Registry: Standardizing the Methods and Practices of Cost-Effectiveness Analysis*. Harvard Center for Risk Analysis, Harvard School of Public Health: Boston, MA. <http://www.hsph.harvard.edu/cearegistry/> or <http://www.hcra.harvard.edu/pdf/preferencescores.pdf> [7 November 2002].
3. Johnson FR, Fries EE, Banzhaf HS. Valuing morbidity: an integration of the willingness-to-pay and health-status index literatures. *J Health Econ* 1997; **16**: 641–665.
4. Cutler DM, Richardson E. *Measuring the health of the US population*. Brookings Papers on Economic Activity: Microeconomics. Brookings Institute: Washington, DC, 1997.
5. Zarkin GA, Dean N, Mauskopf JA, Williams Jr R. Potential health benefits of nutrition label changes. *Am J Public Health* 1993; **83**(5): 717–724.
6. Stanley TD. Wheat from chaff: meta-analysis as quantitative literature review. *J Econ Perspect* 2001; **15**(Summer): 131–150.
7. Smith VK, Pattanayak SK. Is meta-analysis the Noah's Ark for non market valuation? *Environ Resour Econ* 2002; **22**(1–2): 271–296.
8. Smith VK, Osborne LL. Do contingent valuation estimates pass a 'scope' test? A meta-analysis. *J Environ Econ Manage* 1996; **31**(November): 287–301.
9. Hanemann WM. Willingness to pay and willingness to accept: how much can they differ? *Am Econ Rev* 1991; **81**(3): 635–647.
10. Johannsson P-O. *Evaluating Health Risks: An Economic Approach*. Cambridge University Press: Cambridge, MA, 1995.
11. Harrington W, Portney PR. Valuing the benefits of health and safety regulation. *J Urban Econ* 1987; **22**: 101–112.
12. McConnell KE. Does altruism undermine existence value? *J Environ Econ Manage* 1997; **32**: 22–37.
13. Johannsson P-O. Altruism and the value of statistical life. *J Health Econ* 1994; **13**: 111–118.
14. Bergstrom TC. When is a man's life worth more than his human capital? In *The Value of Life and Safety*, Jones-Lee MW (ed.), North-Holland: New York, 1992; 3–26.
15. Johannesson M. *Theory and Methods of Economic Evaluation of Health Care*. Kluwer Academic Publishers: Dordrecht, The Netherlands, 1996.
16. Brazier J, Deverill M, Green C, Harper R, Booth A. A review of the use of health status measures in economic evaluation. *Health Technol Assess* 1999; **3**(9): 1–158.
17. Bleichrodt H, Quiggin J. When is cost-effectiveness analysis equivalent to cost-benefit analysis? *J Health Econ* 1999; **18**: 681–708.
18. Viscusi WK. The value of risks to life and health. *J Econ Lit* 1993; **31**: 1912–1946.
19. Klose T. A utility-theoretic model for QALYs and willingness to pay. *Health Econ* 2003; **12**(1): 17–31.
20. Carson RT. Contingent valuation: a user's guide. *Environ Sci Technol* 2000; **34**: 1413–1418.
21. Smith RD. The discrete-choice willingness-to-pay question format in health economics: should we adopt environmental guidelines? *Med Decis Making* 2000; **20**: 194–206.
22. Blamey RJ, Bennett W, Morrison MD. Yea-saying in contingent valuation surveys. *Land Econ* 1999; **75**: 126–141.
23. Ryan M, Scott DA, Donaldson C. Valuing health care using willingness to pay: a comparison of the payment card and dichotomous choice methods. *J Health Econ* 2004; **23**: 237–258.
24. Frew EJ, Whynes DK, Wolstenholme JL. Eliciting willingness to pay: comparing closed-ended with open-ended and payment scale formats. *Med Decis Making* 2003; **23**(2): 150–159.
25. Holmes TP, Kramer RA. An independent sample test of yea-saying and starting point bias in dichot-